

2.1

(a)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
5	10	0	10	0	8

$$\bar{x} = 1, \bar{y} = 2$$

$$(b) \quad b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = 2 - 0.8 \times 1 = 1.2$$

b_2 : For every one unit increase in x , y is expected to increase by 0.8 units

b_1 : When $x = 0$, the predicted value of y is 1.2

$$(c) \quad \sum x_i^2 = 15, \quad \sum x_i y_i = 18$$

$$\sum x_i^2 - N\bar{x}^2 = 15 - 5 \times 1^2 = 10 = \sum (x_i - \bar{x})^2$$

$$\sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum (x_i - \bar{x})(y_i - \bar{y})$$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
1	10	10	0	3.6	0

$$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{10}{4} = \frac{5}{2}$$

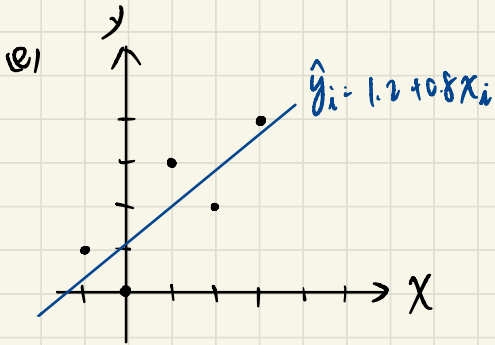
$$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = \frac{10}{4} = \frac{5}{2}$$

$$S_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{8}{\sqrt{\frac{5}{2} \times \frac{5}{2}}} = \frac{2}{\frac{5}{2}} = 0.8$$

$$CV_x = 100 \times \frac{S_x}{\bar{x}} = 100 \times \frac{\sqrt{\frac{5}{2}}}{1} = 158.114$$

Median of X is 1



f) the line passes through the means.

g) $(\bar{x}, \bar{y}) = (1, 2) \Rightarrow \hat{y} = 1.2 + 0.8 \times 1 = 2 = \bar{y}$

h) $\bar{\hat{y}} = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2 = \bar{y}$

i) $\hat{\sigma}^2 = \frac{\sum (y_i - b_1 - b_2 x_i)^2}{N-2} = \frac{\sum e_i^2}{N-2} = \frac{3.6}{3} = 1.2$

j) $\widehat{Var}(b_1 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$\widehat{SE}(b_1) = \sqrt{0.12} = 0.4464$

2.22

(a)

```
Call:
lm(formula = totalscore ~ small, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-191.63  -50.63   -7.12   42.37   324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.628      2.536  362.260  <2e-16 ***
small        9.989       4.611    2.166   0.0305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903, Adjusted R-squared:  0.003071
F-statistic: 4.694 on 1 and 1198 DF, p-value: 0.03047
```

$$\widehat{\text{totalscore}} = 918.628 + 9.989 \times \text{small}$$

當是小班時 (small = 1), totalscore 會
比不是的 (small = 0) 平均多 9.989 分

(b)

```
Call:
lm(formula = readscore ~ small, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-76.123  -21.123   -3.356   15.877   192.877

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  434.123      1.044  416.021  <2e-16 ***
small        5.466       1.897    2.881   0.00403 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882, Adjusted R-squared:  0.006053
F-statistic: 8.301 on 1 and 1198 DF, p-value: 0.004032
```

$$\widehat{\text{readscore}} = 434.123 + 5.466 \times \text{small}$$

當是小班時 (small = 1), readscore 會
比不是的 (small = 0) 平均多 5.466 分

```
Call:
lm(formula = mathscore ~ small, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-145.505  -35.028   -6.505   28.495   141.495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  484.505      1.677  288.884  <2e-16 ***
small        4.522       3.049    1.483   0.138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832, Adjusted R-squared:  0.0009992
F-statistic: 2.199 on 1 and 1198 DF, p-value: 0.1383
```

$$\widehat{\text{mathscore}} = 484.505 + 4.522 \times \text{small}$$

當是小班時 (small = 1), mathscore 會
比不是的 (small = 0) 平均多 4.522 分

(c)

```
Call:
lm(formula = totalscore ~ aide, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-195.14  -52.14   -6.95   40.95   330.86

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  922.145      2.640  349.250  <2e-16 ***
aide        -1.396       4.437   -0.315   0.753
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05, Adjusted R-squared:  -0.000752
F-statistic: 0.09904 on 1 and 1198 DF, p-value: 0.753
```

$$\widehat{\text{totalscore}} = 922.145 - 1.396 \times \text{aide}$$

教師助理與 totalscore 呈負相關
有教師助理的班級 (aide = 1) 會
比沒有的平均 totalscore 少 1.396 分

d1

```
Call:
lm(formula = readscore ~ aide, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-77.908 -21.536  -2.908  15.464 191.092

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  435.9084      1.0882   400.585  <2e-16 ***
aide         -0.3719      1.8285   -0.203    0.839
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05, Adjusted R-squared:  -0.0008002
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389
```

$$\widehat{\text{readscore}} = 435.9084 - 0.3719 \times \text{aide}$$

有教師助理的班級 (aide=1) 會比沒有的平均 readscore 少 0.3719 分

```
Call:
lm(formula = mathscore ~ aide, data = data1)

Residuals:
    Min       1Q   Median       3Q      Max
-147.236 -32.236  -7.212   27.788  140.788

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  486.236      1.744   278.731  <2e-16 ***
aide         -1.024      2.931   -0.349    0.727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019, Adjusted R-squared:  -0.0007327
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268
```

$$\widehat{\text{mathscore}} = 486.236 - 1.024 \times \text{aide}$$

有教師助理的班級 (aide=1) 會比沒有的平均 mathscore 少 1.024 分

2.25

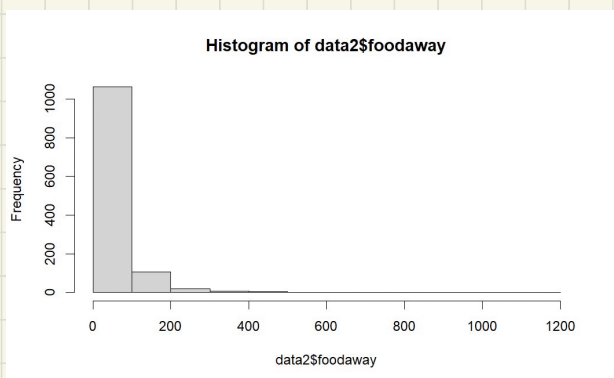
(a)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

mean = 49.29 median = 32.55

25 th percentiles = 12.04

75 th percentiles = 67.50



b)

advanced

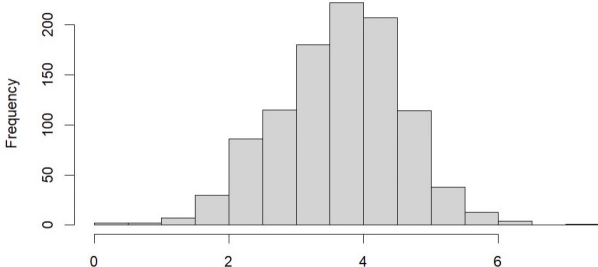
college

no advanced
or college

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	21.67	48.15	73.15	90.00	1179.00
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.44	36.11	48.60	68.67	416.11
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

c)

Histogram of log_foodaway[log_foodaway > 0]



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3646	3.0759	3.6889	3.6547	4.2797	7.0724

$\ln(0)$ 為無定義, 而 foodaway
有些值為 0, 所以那些都不
會被加入樣本.

d)

```
Call:
lm(formula = log(foodaway) ~ income, data = data2[data2$foodaway >
0,])
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-3.6547 -0.5777  0.0530  0.5937  2.7000
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.1293004  0.0565503   55.34  <2e-16 ***
income       0.0069017  0.0006546   10.54  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

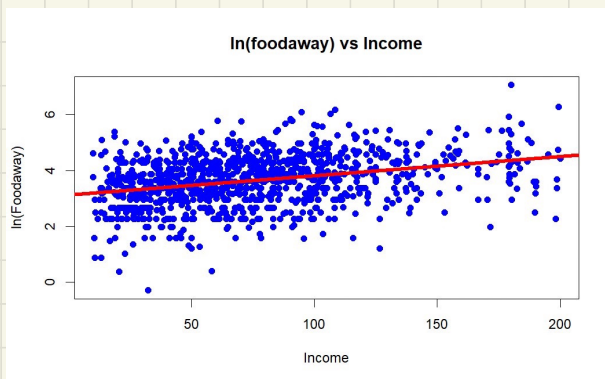
$$\widehat{\ln(\text{foodaway})} = 3.1293004 + 0.0069017 \times \text{income}$$

$$\text{slope} = \frac{dy}{dx} = \frac{d}{dx} (e^{3.1293004 + 0.0069017 \times \text{income}})$$

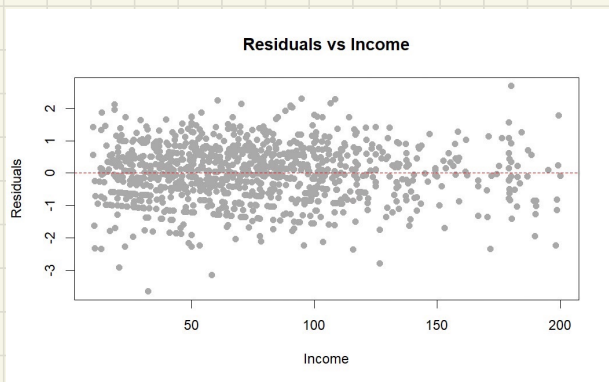
$$= 0.0069017 \times e^{3.1293004 + 0.0069017 \times \text{income}}$$

income 增加 1 單位, foodaway 平均會上升 slope 單位

e)



f)



隨 income 上升, residual 愈來愈稀疏, 可能在模型假設上有錯誤, 可能違反 homoscedasticity 假設